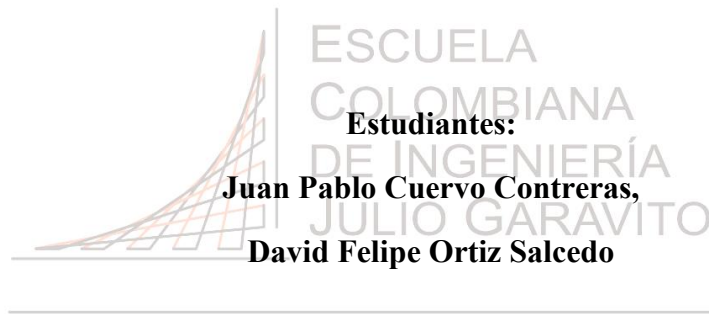


Informe de análisis del proyecto

Universidad Escuela Colombiana de Ingeniería Julio Garavito

Ingeniería de sistemas

Desarrollo Orientado Por Objetos



UNIVERSIDAD

Docentes:

Santiago Rocha Durán,

María Irma Díaz Rozo

16 de abril de 2026

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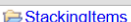


VIGILADA MINEDUCACIÓN

Informe de Análisis Dinámico (Casos de prueba)

Resultado inicial

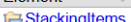
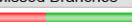
StackingItems (15/04/2026 1:59:31 p. m.)

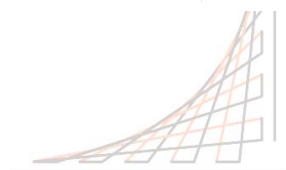
Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
StackingItems		82 %		64 %	103	321	143	852	25	175	2	26
Total	648 of 3.662	82 %	104 of 289	64 %	103	321	143	852	25	175	2	26

A pesar de que ya se cubra más del 75% de las instrucciones del proyecto Stacking Cups donde los casos de prueba cubren el 82% del proyecto, se decidió agregar un caso de prueba adicional en el paquete Tower con el fin de asegurar un poco más los métodos que aún no están probados, y así asegurar ciertos casos de error no vistos aún en el simulador que pueden suceder.

Resultado final

StackingItems (15/04/2026 2:19:02 p. m.)

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
StackingItems		82 %		64 %	103	322	143	857	25	176	2	26
Total	648 of 3.680	82 %	104 of 289	64 %	103	322	143	857	25	176	2	26



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Informe de Análisis Estático (PMD)

Resultado inicial

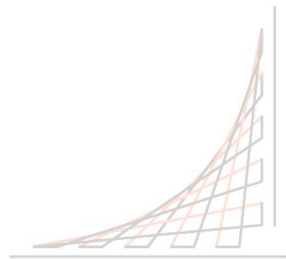
PMD report			
Problems found			
#	File	Line	Problem
1	src/module-info.java	1	DanglingJavadoc: Javadoc comment does not belong to any class, method or field
2	src/shapes/Canvas.java	8	CommentSize: Comment is too large: Too many lines
3	src/shapes/Canvas.java	18	ClassWithOnlyPrivateConstructorsShouldBeFinal: This class has only private constructors and may be final
4	src/shapes/Canvas.java	24	CommentRequired: Field comments are required
5	src/shapes/Canvas.java	30	NonThreadSafeSingleton: Singleton is not thread safe
6	src/shapes/Canvas.java	40	CommentRequired: Field comments are required
7	src/shapes/Canvas.java	40	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
8	src/shapes/Canvas.java	40	ImmutableField: Field 'frame' may be declared final
9	src/shapes/Canvas.java	41	AvoidFieldNameMatchingTypeName: It is somewhat confusing to have a field name matching the declaring class name
10	src/shapes/Canvas.java	41	CommentRequired: Field comments are required
11	src/shapes/Canvas.java	41	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
12	src/shapes/Canvas.java	41	ImmutableField: Field 'canvas' may be declared final
13	src/shapes/Canvas.java	42	CommentRequired: Field comments are required
14	src/shapes/Canvas.java	42	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
15	src/shapes/Canvas.java	43	CommentRequired: Field comments are required
16	src/shapes/Canvas.java	43	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
17	src/shapes/Canvas.java	43	ImmutableField: Field 'backgroundColour' may be declared final
18	src/shapes/Canvas.java	44	CommentRequired: Field comments are required
19	src/shapes/Canvas.java	44	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
20	src/shapes/Canvas.java	45	CommentRequired: Field comments are required
21	src/shapes/Canvas.java	45	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
22	src/shapes/Canvas.java	45	ImmutableField: Field 'objects' may be declared final
23	src/shapes/Canvas.java	46	CommentRequired: Field comments are required
24	src/shapes/Canvas.java	46	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.

Se obtuvieron 712 errores de acuerdo con el formato PMD, sin embargo, la mayoría de estos errores era sobre todo fallos en sintaxis que la aplicación de Eclipse nos aconsejaba corregir y errores muy mínimos sobre una mejor práctica en la programación o métodos que deberían retornar listas en vez de valores nulos, entre otros.

Resultado final

PMD report			
Problems found			
#	File	Line	Problem
1	src/module-info.java	1	DanglingJavadoc: Javadoc comment does not belong to any class, method or field
2	src/shapes/Canvas.java	8	CommentSize: Comment is too large: Too many lines
3	src/shapes/Canvas.java	24	CommentRequired: Field comments are required
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22	src/shapes/Canvas.java	46	CommentRequired: Field comments are required
23	src/shapes/Canvas.java	46	FieldDeclarationsShouldBeAtStartOfClass: Fields should be declared at the top of the class, before any method declarations, constructors, initializers or inner classes.
24	src/shapes/Canvas.java	46	ImmutableField: Field 'shapes' may be declared final

380 src/tower/TowerContestTest.java	45	JUnit5AssertionsShouldIncludeMessage: Unit test assertions should include a message
381 src/tower/TowerContestTest.java	50	CommentSize: Comment is too large: Line too long
382 src/tower/TowerContestTest.java	53	JUnit5TestShouldBePackagePrivate: JUnit 5 tests should be package-private
383 src/tower/TowerContestTest.java	63	JUnit5TestShouldBePackagePrivate: JUnit 5 tests should be package-private
384 src/tower/TowerContestTest.java	74	JUnit5TestShouldBePackagePrivate: JUnit 5 tests should be package-private
385 src/tower/TowerContestTest.java	84	JUnit5TestShouldBePackagePrivate: JUnit 5 tests should be package-private
386 src/tower/TowerContestTest.java	84	JUnitTestContainsTooManyAsserts: Unit tests should not contain more than 1 assert(s)
387 src/tower/TowerContestTest.java	87	CloseResource: Ensure that resources like this PrintStream object are closed after use
388 src/tower/TowerContestTest.java	88	RelianceOnDefaultCharset: Specify a character set instead of relying on the default charset
389 src/tower/TowerContestTest.java	97	RelianceOnDefaultCharset: Specify a character set instead of relying on the default charset
390 src/tower/TowerContestTest.java	116	SystemPrintln: Usage of System.out.err
391 src/tower/TowerContestTest.java	121	SystemPrintln: Usage of System.out.err
392 src/tower/TowerContestTest.java	129	UncommentedEmptyMethodBody: Document empty method body



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